

Kelompok 8

Skenario Database pada Postgresql

Test Tabel Pemain

The screenshot shows a PostgreSQL client interface with the following components:

- Query Editor:** Contains SQL queries for inserting data into the `atribut_stat` table based on character names.
- Data Output:** Displays the results of the queries, showing three rows of data.
- Messages and Notifications:** Empty sections for error messages or system notifications.

SQL Queries:

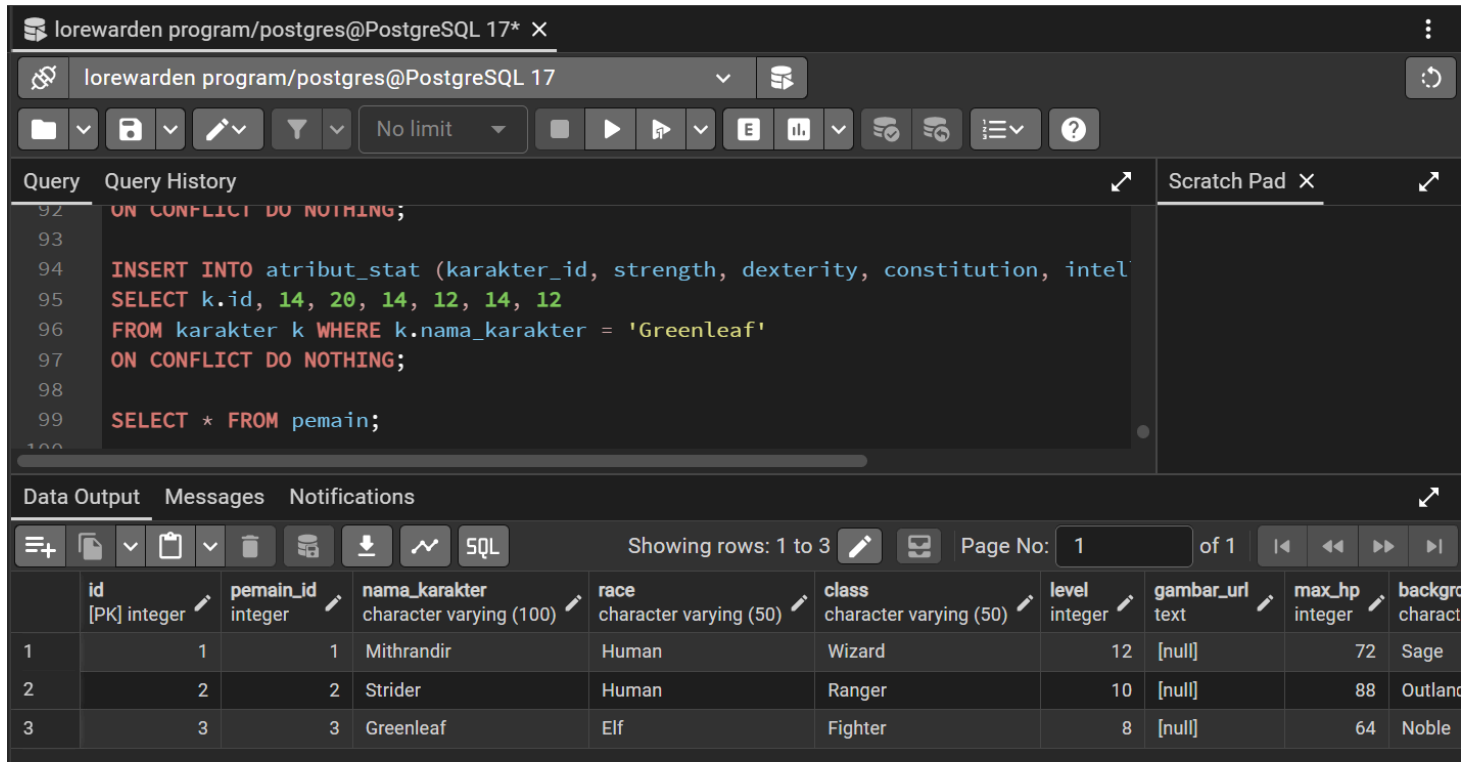
```
86 FROM karakter k WHERE k.nama_karakter = 'Mithrandir'
87 ON CONFLICT DO NOTHING;
88
89 INSERT INTO atribut_stat (karakter_id, strength, dexterity, constitution, intell
90 SELECT k.id, 18, 16, 16, 14, 14, 16
91 FROM karakter k WHERE k.nama_karakter = 'Strider'
92 ON CONFLICT DO NOTHING;
93
94 INSERT INTO atribut_stat (karakter_id, strength, dexterity, constitution, intell
95 SELECT k.id, 14, 20, 14, 12, 14, 12
96 FROM karakter k WHERE k.nama_karakter = 'Greenleaf'
97 ON CONFLICT DO NOTHING;
98
```

Data Output:

	id [PK] integer	username character varying (100)	discord_id character varying (50)	created_at timestamp without time zone
1	1	Gandalf_the_Grey	gandalf#1234	2026-05-17 21:30:48.683507
2	2	Aragorn_II	aragorn#5678	2026-05-17 21:30:48.683507
3	3	Legolas_Greenleaf	legolas#9012	2026-05-17 21:30:48.683507

- Hasil pada PostgreSQL menunjukkan bahwa database lorewarden telah berhasil dibuat dan dijalankan dengan baik. Tabel pemain berhasil menyimpan data pengguna seperti username, discord_id, dan created_at yang menandakan waktu data dibuat. Data yang tampil, seperti Gandalf_the_Grey, Aragorn_II, dan Legolas_Greenleaf, merupakan seed data awal yang dimasukkan melalui file init.sql.
- Dimana ini membuktikan bahwa proses pembuatan tabel, insert data, serta konfigurasi database pada PostgreSQL telah berjalan dengan sukses dan database siap digunakan untuk pengujian relasi maupun query lanjutan.

Test Tabel Karakter

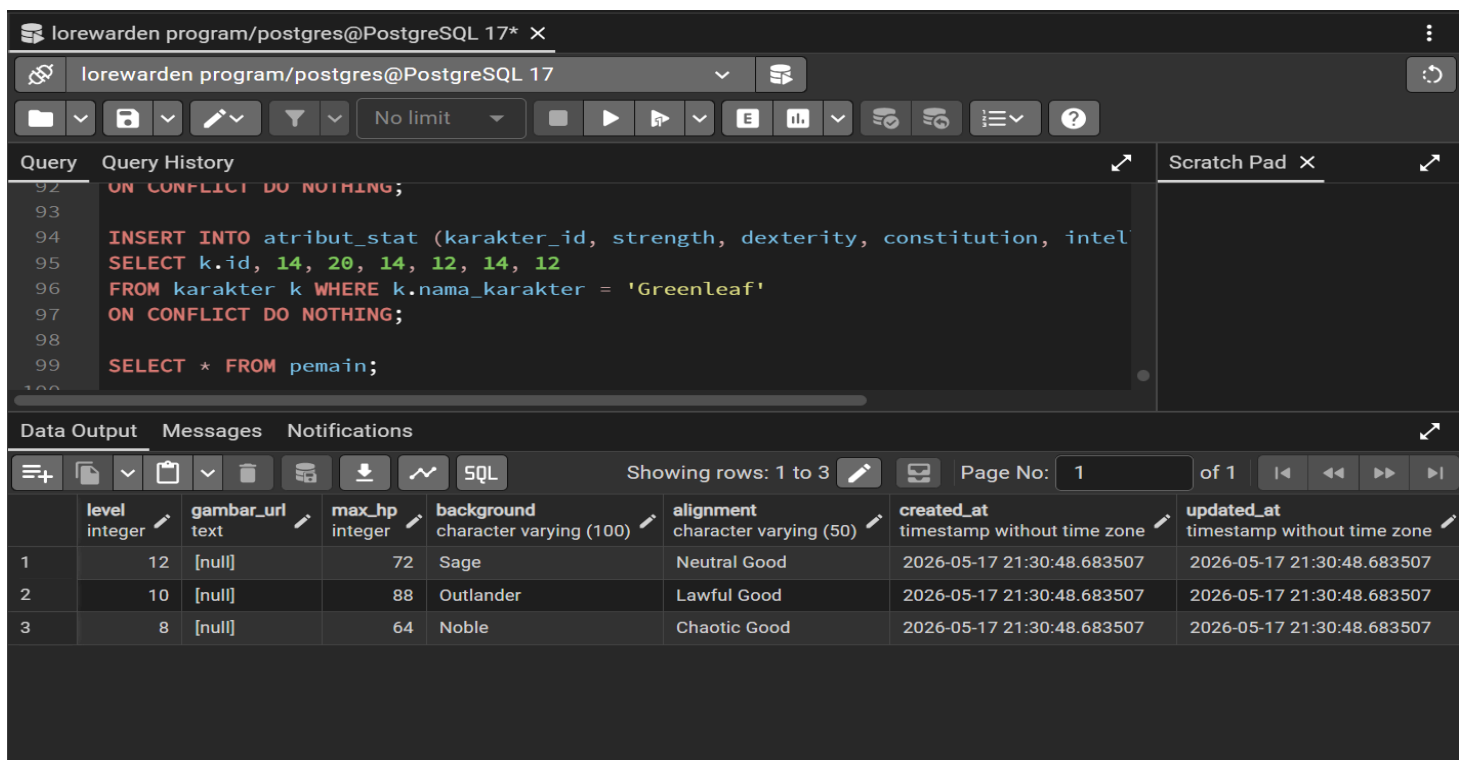


The screenshot shows a PostgreSQL query editor window titled "lorewarden program/postgres@PostgreSQL 17* X". The query editor contains the following SQL code:

```
92 ON CONFLICT DO NOTHING;
93
94 INSERT INTO atribut_stat (karakter_id, strength, dexterity, constitution, intel
95 SELECT k.id, 14, 20, 14, 12, 14, 12
96 FROM karakter k WHERE k.nama_karakter = 'Greenleaf'
97 ON CONFLICT DO NOTHING;
98
99 SELECT * FROM pemain;
```

The "Data Output" tab shows the result of the query, displaying a table with 10 columns: id, pemain_id, nama_karakter, race, class, level, gambar_url, max_hp, and background. The table contains 3 rows of data:

	id [PK] integer	pemain_id integer	nama_karakter character varying (100)	race character varying (50)	class character varying (50)	level integer	gambar_url text	max_hp integer	background character varying (100)
1	1	1	Mithrandir	Human	Wizard	12	[null]	72	Sage
2	2	2	Strider	Human	Ranger	10	[null]	88	Outland
3	3	3	Greenleaf	Elf	Fighter	8	[null]	64	Noble



The screenshot shows a PostgreSQL query editor window titled "lorewarden program/postgres@PostgreSQL 17* X". The query editor contains the following SQL code:

```
92 ON CONFLICT DO NOTHING;
93
94 INSERT INTO atribut_stat (karakter_id, strength, dexterity, constitution, intel
95 SELECT k.id, 14, 20, 14, 12, 14, 12
96 FROM karakter k WHERE k.nama_karakter = 'Greenleaf'
97 ON CONFLICT DO NOTHING;
98
99 SELECT * FROM pemain;
```

The "Data Output" tab shows the result of the query, displaying a table with 8 columns: level, gambar_url, max_hp, background, alignment, created_at, and updated_at. The table contains 3 rows of data:

	level integer	gambar_url text	max_hp integer	background character varying (100)	alignment character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	12	[null]	72	Sage	Neutral Good	2026-05-17 21:30:48.683507	2026-05-17 21:30:48.683507
2	10	[null]	88	Outlander	Lawful Good	2026-05-17 21:30:48.683507	2026-05-17 21:30:48.683507
3	8	[null]	64	Noble	Chaotic Good	2026-05-17 21:30:48.683507	2026-05-17 21:30:48.683507

- Hasil query pada tabel karakter menunjukkan bahwa sistem database berhasil menyimpan data karakter yang terhubung dengan pemain melalui atribut pemain_id. Setiap karakter memiliki informasi lengkap seperti nama karakter, ras, class, level, maksimum HP, background, alignment, serta timestamp created_at dan updated_at.
- Data seperti Mithrandir, Strider, dan Greenleaf membuktikan bahwa relasi antar tabel berjalan dengan baik dan PostgreSQL mampu mengelola data karakter RPG secara terstruktur menggunakan primary key dan foreign key untuk menjaga integritas database.

Test Relasi Database

The screenshot shows a PostgreSQL client window titled 'lorewarden program/postgres@PostgreSQL 17*'. The interface includes a toolbar with icons for file operations, query execution, and settings. The 'Query' tab is active, displaying the following SQL query:

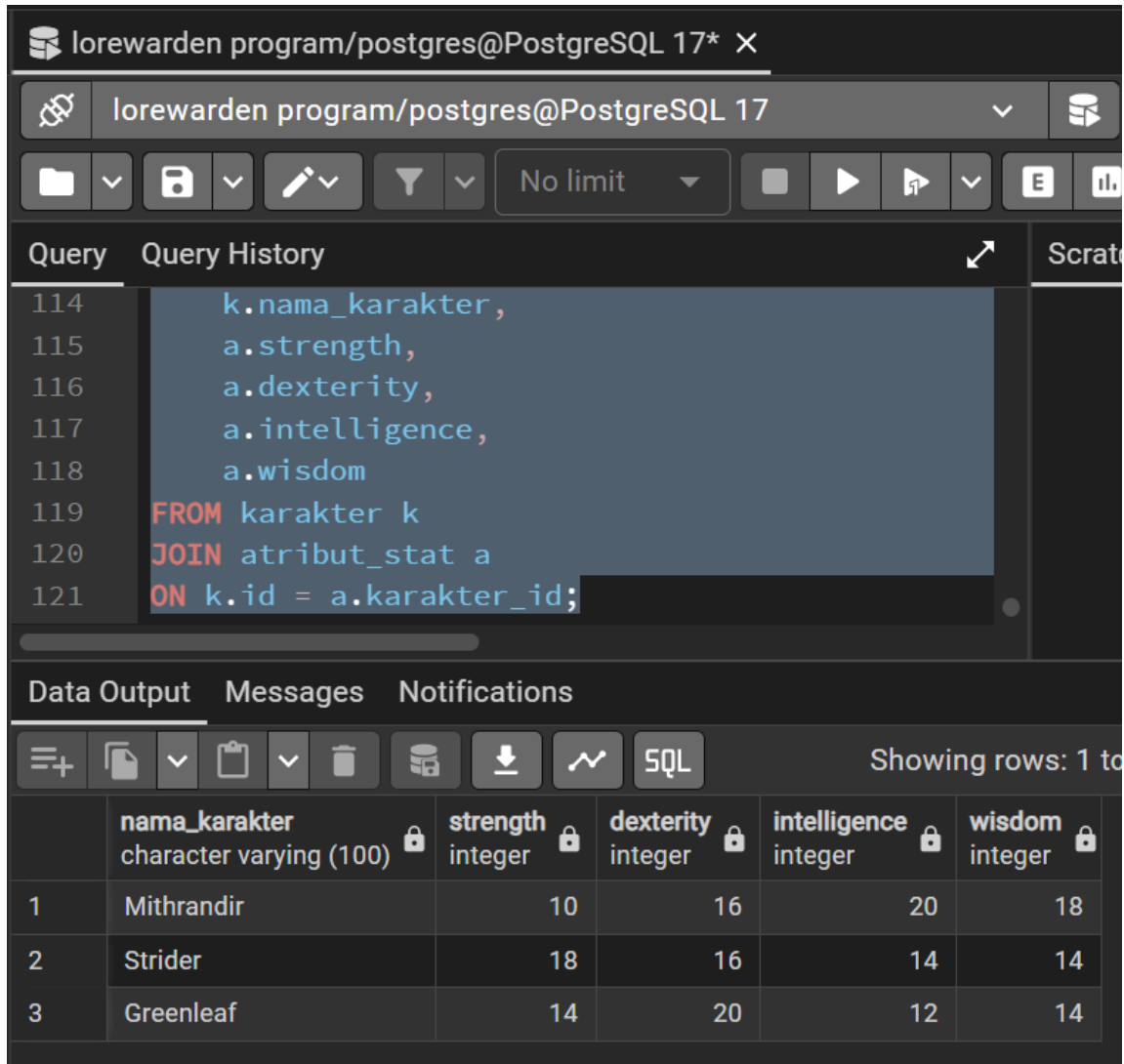
```
99 SELECT * FROM pemain;  
100  
101 SELECT * FROM karakter;  
102  
103 SELECT  
104     p.username,  
105     k.nama_karakter,  
106     k.race,
```

The 'Data Output' tab is also visible, showing the results of the query. The results are displayed in a table with 6 columns: username, nama_karakter, race, class, and level. The table contains 3 rows of data.

	username character varying (100)	nama_karakter character varying (100)	race character varying (50)	class character varying (50)	level integer
1	Gandalf_the_Grey	Mithrandir	Human	Wizard	12
2	Aragorn_II	Strider	Human	Ranger	10
3	Legolas_Greenleaf	Greenleaf	Elf	Fighter	8

- Hasil query JOIN tersebut menunjukkan bahwa relasi antara tabel pemain dan karakter berhasil diimplementasikan menggunakan foreign key pemain_id. Query menampilkan data gabungan berupa username pemain beserta informasi karakter yang dimiliki, seperti nama karakter, ras, class, dan level. Data yang muncul membuktikan bahwa PostgreSQL mampu menghubungkan data antar tabel secara konsisten sehingga informasi dapat diakses secara lebih terstruktur dan efisien dalam satu query.

TEST ATRIBUT STAT



The screenshot shows a PostgreSQL IDE interface. The top bar indicates the connection is to 'lorewarden program/postgres@PostgreSQL 17*'. Below the toolbar, the 'Query' tab is active, displaying a SQL query. The query is as follows:

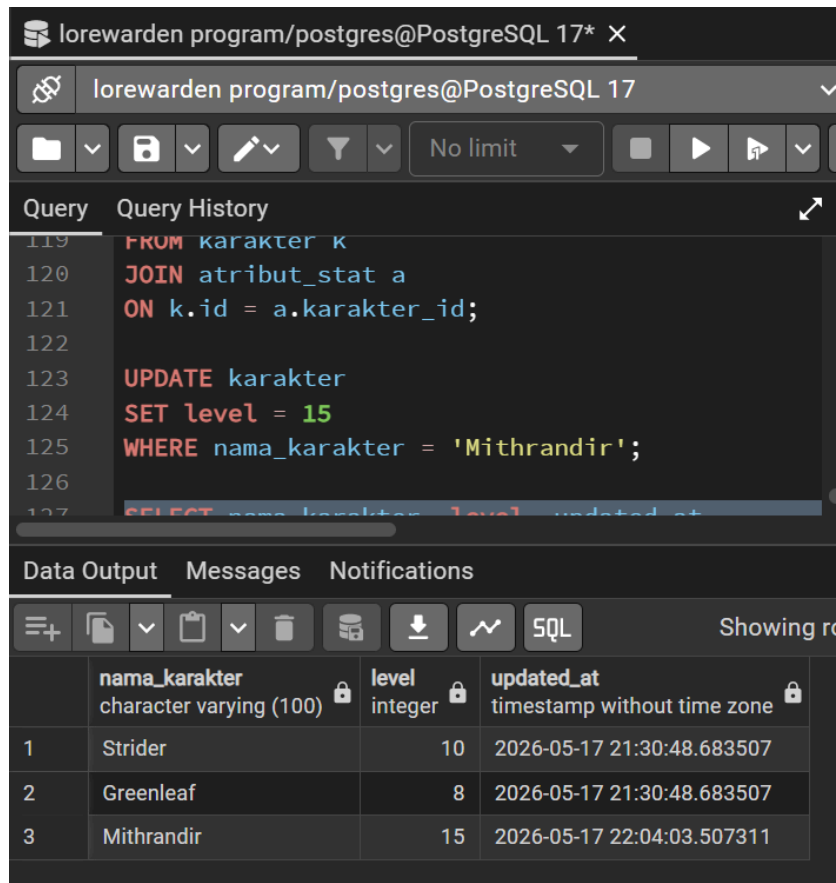
```
114     k.nama_karakter,  
115     a.strength,  
116     a.dexterity,  
117     a.intelligence,  
118     a.wisdom  
119 FROM karakter k  
120 JOIN atribut_stat a  
121 ON k.id = a.karakter_id;
```

Below the query editor, the 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with 6 columns: nama_karakter, strength, dexterity, intelligence, and wisdom. The table shows 3 rows of data:

	nama_karakter character varying (100)	strength integer	dexterity integer	intelligence integer	wisdom integer
1	Mithrandir	10	16	20	18
2	Strider	18	16	14	14
3	Greenleaf	14	20	12	14

- Hasil query tersebut menunjukkan bahwa tabel atribut_stat berhasil terhubung dengan tabel karakter melalui relasi foreign key karakter_id. Query menampilkan atribut utama setiap karakter seperti strength, dexterity, intelligence, dan wisdom sehingga sistem dapat menyimpan statistik karakter secara terpisah namun tetap terintegrasi dengan data karakter utama. Implementasi ini menunjukkan bahwa PostgreSQL mampu mengelola data relasional secara terstruktur dan mendukung pengolahan informasi karakter RPG dengan lebih efisien.

Cek updated_at



The screenshot shows a PostgreSQL client window titled 'lorewarden program/postgres@PostgreSQL 17*'. The query editor contains the following SQL code:

```
119 FROM karakter k
120 JOIN atribut_stat a
121 ON k.id = a.karakter_id;
122
123 UPDATE karakter
124 SET level = 15
125 WHERE nama_karakter = 'Mithrandir';
126
127 SELECT nama_karakter, level, updated_at
```

The 'Data Output' tab is active, displaying the results of the query. The table has four columns: 'nama_karakter' (character varying (100)), 'level' (integer), and 'updated_at' (timestamp without time zone). The results show three rows of data:

	nama_karakter character varying (100)	level integer	updated_at timestamp without time zone
1	Strider	10	2026-05-17 21:30:48.683507
2	Greenleaf	8	2026-05-17 21:30:48.683507
3	Mithrandir	15	2026-05-17 22:04:03.507311

- Hasil query tersebut menunjukkan bahwa proses UPDATE pada tabel karakter berhasil dijalankan, di mana level karakter Mithrandir berubah dari level sebelumnya menjadi level 15. Perubahan pada kolom updated_at membuktikan bahwa trigger otomatis pada PostgreSQL bekerja dengan baik untuk mencatat waktu terakhir data diperbarui. Implementasi ini membantu menjaga konsistensi dan pelacakan perubahan data dalam sistem database Lorewarden.